Now that the scope is finalized, I would strongly recommend treating this as a **3-layer project**:
Layer 1 → Research & AI Core (70%)
Layer 2 → Industry Features (20%)
Layer 3 → UI & Deployment (10%)
For publication and interviews, the AI core matters far more than the frontend.

Overall Timeline
Assuming:

- Mini Project Duration: 3–4 months
- Working ~2–3 hours/day

Month 1
Research + Dataset + Baseline Models
Month 2
Advanced Models + Explainability + Severity Scoring
Month 3
Similarity Search + Active Learning + Drift Detection
Month 4
Dashboard + Deployment + Paper Writing

Phase 1: Research & Literature Survey (Week 1)
Goal
Understand existing industrial anomaly detection methods.
Read

1. MVTec AD Paper
2. PatchCore Paper
3. PaDiM Paper
4. EfficientAD Paper
5. FastFlow Paper

Deliverables
Create:
literature_review.md

Existing Methods
Limitations
Research Gap
Proposed Solution

Phase 2: Dataset Analysis (Week 1)
Dataset
MVTec AD
Categories:
Bottle
Cable
Capsule
Grid
Hazelnut
Leather
Metal Nut

Pill
Screw
Tile
Toothbrush
Transistor
Wood
Zipper

Analyze

For each category:
Normal Images
Defect Types
Ground Truth Masks

Deliverable

Dataset Report

Phase 3: Data Pipeline (Week 2)

Build Preprocessing Pipeline

Operations

Resize
Normalize
Augment

Augmentations

Rotation
Brightness
Contrast
Noise
Flip
Libraries:
Albumentations
OpenCV

Deliverable

Reusable Dataset Loader

Phase 4: Baseline Model (Week 2)

EfficientNet-B0

Task:
Normal
vs
Defective

Metrics

- Accuracy
- Precision
- Recall
- F1

Deliverable

Baseline Classification Model

Phase 5: Explainability (Week 3)

Grad-CAM
Generate:
Prediction
Confidence
Heatmap
Example:
Defective

Confidence:
97%

Defect Area:
Highlighted
Deliverable
Visual Explanation Module

Phase 6: Autoencoder Anomaly Detection (Week 3)
Train only on:
Normal Images
Output
Reconstruction Error
Generate

- Anomaly Heatmap
- Localization Map

Deliverable
Unsupervised Detection Module

Phase 7: Research Models (Week 4–5)
Implement:
PaDiM
Then:
PatchCore
Compare against:
EfficientNet
Autoencoder
PaDiM
PatchCore
Metrics

- AUROC
- F1
- Inference Time

Deliverable
Research Benchmark

Phase 8: Severity Scoring (Week 5)
Create:
Severity Score
0–100

Based on:
Anomaly Score
Defect Area
Confidence
Output
Minor
Moderate
Critical
Deliverable
Severity Engine

Phase 9: Similarity Search (Week 6)
FAISS
Generate embeddings:
EfficientNet Features
Store:
Vector Database
Output
Top 5 Similar Defects
Deliverable
Defect Retrieval Module

Phase 10: Active Learning (Week 6)
Detect uncertain predictions:
Confidence < Threshold
Ask user:
Verify Prediction
Store feedback.
Deliverable
Active Learning Pipeline

Phase 11: Drift Detection (Week 7)
Monitor:
Training Distribution
Production Distribution
Methods:

- KL Divergence
- Embedding Distance

Output
Data Drift Alert
Deliverable
Production Monitoring Module

Phase 12: Batch Inspection (Week 7)
User uploads:
100+
500+
1000+

Images
System processes all.
Output:
Defective Count
Severity Distribution
Defect Rate
Deliverable
Batch Inspection Engine

Phase 13: Folder Monitoring (Week 8)
Create:
watch_folder/
System:
Watch
Detect
Store
Update Dashboard
Use:
watchdog
Deliverable
Continuous Inspection Simulator

Phase 14: Live Camera Inspection (Week 8)
Input:
Webcam
Pipeline:
Capture
Inference
Heatmap
Display
Libraries:
OpenCV
Deliverable
Real-Time Inspection Module

Phase 15: Smart Factory Dashboard (Week 9)
Metrics
Show:
Today's Inspections
Defect Rate
Critical Defects
Most Common Defect
Charts
Defect Trend
Severity Trend
Model Performance
Deliverable
Analytics Dashboard

Phase 16: FastAPI Backend (Week 9)
Endpoints:
/predict
/heatmap
/batch
/similarity
/feedback
/report
/drift
Deliverable
Production API

Phase 17: Database (Week 9)
Use:
PostgreSQL
Tables:
users
inspections
predictions
feedback
reports
alerts
Deliverable
Persistent Storage

Phase 18: React Frontend (Week 10)
Pages:
Home
Upload Inspection
Batch Inspection
Live Monitoring
Dashboard
Reports
Model Comparison
Admin Panel
Deliverable
Complete UI

Phase 19: Report Generation (Week 10)
Generate:
PDF
Contains:
Image
Prediction
Severity
Heatmap
Confidence

Similar Defects

Deliverable

Automated Reports

Phase 20: Deployment (Week 11)

Frontend:

- Vercel

Backend:

- Render

Database:

- Supabase

Models:

- Hugging Face Spaces

Phase 21: Research Paper (Week 12)

Proposed Contributions

1. Hybrid Defect Detection Framework
2. Explainable AI for Inspection
3. Severity-Aware Quality Assessment
4. Active Learning Feedback Loop
5. Similarity-Based Defect Retrieval
6. Continuous Inspection Pipeline
7. Drift-Aware Manufacturing Monitoring

Target Publication Venues

For a student mini-project, aim for:

- IEEE conferences
- Springer conference proceedings
- Elsevier journals (if results are strong)

What You Should Build First

Month 1 (Mandatory)

- ✅ Dataset Pipeline
- ✅ EfficientNet
- ✅ Grad-CAM
- ✅ Autoencoder

Month 2 (Research Core)

- ✅ PatchCore
- ✅ PaDiM
- ✅ Severity Scoring

Month 3 (Innovation)

- ✅ Similarity Search
- ✅ Active Learning
- ✅ Drift Detection

Month 4 (Productization)

- ✅ Batch Inspection
- ✅ Folder Monitoring

✅ Live Camera
✅ Dashboard
✅ Deployment

This order minimizes risk and ensures that even if time runs short, you will already have the strongest AI/ML components completed. The research contributions (PatchCore + explainability + severity scoring + active learning + similarity search) are what will matter most for interviews and publication potential.